KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4241 | 4 Credits: 4 | Program Core | Source-grounded

Therapeutic Equipment

● Introduce the different types of therapeutic equipment used in the medical field.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4241 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4241.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Biomedical Engineering
Course code	4241
Course title	Therapeutic Equipment
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

14 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● Introduce the different types of therapeutic equipment used in the medical field.
- ● Impart a thorough knowledge in the principle and operation of ICU and Operation theatre equipment
- ● Give an account of the principle, working and applications of surgical instruments.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding defibrillators and electrical stimulators. Explain the principle and working of instruments	See official syllabus
CO2	13 Understanding for surgery, Dialysis and lithotripsy. Summarize the principle and operation of the ICU	See official syllabus
CO3	15 Understanding and operation theatre equipment. Describe the working of drug delivery systems,	See official syllabus
CO4	15 Understanding endoscopy systems and radiotherapy unit.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2

Row	Extracted official mapping
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	stimulators.	3	As specified
Module 2	lithotripsy.	3	As specified
Module 3	equipment.	4	As specified
Module 4	radiotherapy unit.	4	As specified

MODULE 1

stimulators.

This module is presented from the official syllabus scope for Therapeutic Equipment. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: stimulators.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 6 Understanding principle and working Identify the types of defibrillators, their

6 Understanding principle and working Identify the types of defibrillators, their is an official topic in Module 1 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding principle and working Identify the types of defibrillators, their using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding principle and working identify the types of defibrillators, their should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 6 Understanding principle and working Identify the types of defibrillators, their means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 6 Understanding principle and working Identify the types of defibrillators, their □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ 6 Understanding principle and working. Explain pain relief through electrical stimulation

6 Understanding principle and working. Explain pain relief through electrical stimulation is an official topic in Module 1 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding principle and working. Explain pain relief through electrical stimulation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding principle and working. explain pain relief through electrical stimulation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 6 Understanding principle and working. Explain pain relief through electrical stimulation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 6 Understanding principle and working. Explain pain relief through electrical stimulation □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ **3 Understanding and types of electrical stimulators. Cardiac pacemakers - design requirements - Power sources - Pacing modes - Working of ventricular synchronous demand pacemaker -Programmable pacemaker (block diagram and description). Defibrillators - Principle - working and output waveforms of DC defibrillators - Defibrillator with synchronizer - cardioverter -Automatic Implantable Defibrillators (block diagram and working) Electrical stimulators - Principle - different types of electrical stimulators - TENS- Spinal Cord Stimulator - Magnetic stimulation Explain the principle and working of instruments for surgery, Dialysis and**

3 Understanding and types of electrical stimulators. Cardiac pacemakers - design requirements - Power sources - Pacing modes - Working of ventricular synchronous demand pacemaker -Programmable pacemaker (block diagram and description). Defibrillators - Principle - working and output waveforms of DC defibrillators - Defibrillator with synchronizer - cardioverter -Automatic Implantable Defibrillators (block diagram and working) Electrical stimulators - Principle - different types of electrical stimulators - TENS- Spinal Cord Stimulator - Magnetic stimulation Explain the principle and working of instruments for surgery, Dialysis and is an official topic in Module 1 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and types of electrical stimulators. Cardiac pacemakers - design requirements - Power sources - Pacing modes -

Working of ventricular synchronous demand pacemaker -Programmable pacemaker (block diagram and description). Defibrillators - Principle - working and output waveforms of DC defibrillators - Defibrillator with synchronizer - cardioverter -Automatic Implantable Defibrillators (block diagram and working) Electrical stimulators - Principle - different types of electrical stimulators - TENS- Spinal Cord Stimulator - Magnetic stimulation Explain the principle and working of instruments for surgery, Dialysis and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and types of electrical stimulators. cardiac pacemakers - design requirements - power sources - pacing modes - working of ventricular synchronous demand pacemaker -programmable pacemaker (block diagram and description). defibrillators - principle - working and output waveforms of dc defibrillators - defibrillator with synchronizer - cardioverter -automatic implantable defibrillators (block diagram and working) electrical stimulators - principle - different types of electrical stimulators - tens- spinal cord stimulator - magnetic stimulation explain the principle and working of instruments for surgery, dialysis and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding and types of electrical stimulators. Cardiac pacemakers - design requirements - Power sources - Pacing modes - Working of ventricular synchronous demand pacemaker -Programmable pacemaker (block diagram and description). Defibrillators - Principle - working and output waveforms of DC defibrillators - Defibrillator with synchronizer - cardioverter -Automatic Implantable Defibrillators (block diagram and working) Electrical stimulators - Principle - different types

Aspect	What to prepare
	of electrical stimulators - TENS- Spinal Cord Stimulator - Magnetic stimulation Explain the principle and working of instruments for surgery, Dialysis and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding and types of electrical stimulators. Cardiac pacemakers - design requirements - Power sources - Pacing modes - Working of ventricular synchronous demand pacemaker - Programmable pacemaker (block diagram and description). Defibrillators - Principle - working and output waveforms of DC defibrillators - Defibrillator with synchronizer - cardioverter -Automatic Implantable Defibrillators (block diagram and working) Electrical stimulators - Principle - different types of electrical stimulators - TENS- Spinal Cord Stimulator - Magnetic stimulation Explain the principle and working of instruments for surgery, Dialysis and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

lithotripsy.

This module is presented from the official syllabus scope for Therapeutic Equipment. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: lithotripsy.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding diathermy unit. Classify Lithotripters based on their principle

4 Understanding diathermy unit. Classify Lithotripters based on their principle is an official topic in Module 2 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding diathermy unit. Classify Lithotripters based on their principle using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding diathermy unit. classify lithotripters based on their principle should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding diathermy unit. Classify Lithotripters based on their principle means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding diathermy unit. Classify Lithotripters based on their principle □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ **and working. Describe dialysis and the working of**

and working. Describe dialysis and the working of is an official topic in Module 2 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and working. Describe dialysis and the working of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and working. describe dialysis and the working of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What and working. Describe dialysis and the working of means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ and working. Describe dialysis and the working of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ **4 Understanding heamodialyzers Surgical diathermy -Working principle - modes of operation(schematic diagram and waveforms) - block diagram and working of ESU Lithotripsy - Principle - operation of extra corporeal shock wave lithotripter and ultrasonic lithotripter with schematic Dialysis - Principle - types of dialyzers (parallel flow, hollow fibre and coil type) - heamodialysis machine principle and working with block diagram Summarize the principle and operation of the ICU and operation theatre**

4 Understanding heamodialyzers Surgical diathermy -Working principle - modes of operation(schematic diagram and waveforms) - block diagram and working of ESU Lithotripsy - Principle - operation of extra corporeal shock wave lithotripter and ultrasonic lithotripter with schematic Dialysis - Principle - types of dialyzers (parallel flow, hollow fibre and coil type) - heamodialysis machine principle and working with block diagram Summarize the principle and operation of the ICU and operation theatre is an official topic in Module 2 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding heamodialyzers Surgical diathermy -Working principle - modes of operation(schematic diagram and waveforms) - block diagram and working of ESU Lithotripsy - Principle - operation of extra corporeal shock wave lithotripter and ultrasonic lithotripter with schematic Dialysis - Principle - types of dialyzers (parallel flow, hollow fibre and coil type) - heamodialysis machine principle and working with block

diagram Summarize the principle and operation of the ICU and operation theatre using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding heamodialyzers surgical diathermy - working principle - modes of operation(schematic diagram and waveforms) - block diagram and working of esu lithotripsy - principle - operation of extra corporeal shock wave lithotripter and ultrasonic lithotripter with schematic dialysis - principle - types of dialyzers (parallel flow, hollow fibre and coil type) - heamodialysis machine principle and working with block diagram summarize the principle and operation of the icu and operation theatre should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding heamodialyzers Surgical diathermy -Working principle - modes of operation(schematic diagram and waveforms) - block diagram and working of ESU Lithotripsy - Principle - operation of extra corporeal shock wave lithotripter and ultrasonic lithotripter with schematic Dialysis - Principle - types of dialyzers (parallel flow, hollow fibre and coil type) - heamodialysis machine principle and working with block diagram Summarize the principle and operation of the ICU and operation theatre means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding heamodialyzers Surgical diathermy -Working principle - modes of operation(schematic diagram and waveforms) - block diagram and working of ESU Lithotripsy - Principle - operation of extra corporeal shock wave lithotripter and ultrasonic lithotripter with schematic Dialysis - Principle - types of dialyzers (parallel flow, hollow fibre and coil type) - heamodialysis machine principle and working with block diagram Summarize the principle and operation of the ICU and operation theatre □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

equipment.

This module is presented from the official syllabus scope for Therapeutic Equipment. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: equipment.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 6 Understanding Respiration and the types of ventilators. Explain the principle and working of

6 Understanding Respiration and the types of ventilators. Explain the principle and working of is an official topic in Module 3 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding Respiration and the types of ventilators. Explain the principle and working of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding respiration and the types of ventilators. explain the principle and working of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 6 Understanding Respiration and the types of ventilators. Explain the principle and working of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 6 Understanding Respiration and the types of ventilators. Explain the principle and working of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ anesthesia machine

anesthesia machine is an official topic in Module 3 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify anesthesia machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, anesthesia machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What anesthesia machine means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ anesthesia machine □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Identify the need for cardiopulmonary bypass. 2 Understanding Describe the principle working of Heart lung

Identify the need for cardiopulmonary bypass. 2 Understanding Describe the principle working of Heart lung is an official topic in Module 3 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the need for cardiopulmonary bypass. 2 Understanding Describe the principle working of Heart lung using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the need for cardiopulmonary bypass. 2 understanding describe the principle working of heart lung should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the need for cardiopulmonary bypass. 2 Understanding Describe the principle working of Heart lung means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Identify the need for cardiopulmonary bypass. 2 Understanding Describe the principle working of Heart lung □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ 3 Understanding machine Ventilators - terminologies - cycling mechanisms - modes of Operation - working of a typical ventilator (block diagram). Anaesthesia machine - principle - block diagram and working Heart lung machine - Principle and need of cardiopulmonary bypass - Operation and types of oxygenators (bubble, film and membrane type) - block diagram Describe the working of drug delivery systems, endoscopy systems and

3 Understanding machine Ventilators - terminologies - cycling mechanisms - modes of Operation - working of a typical ventilator (block diagram). Anaesthesia machine - principle - block diagram and working Heart lung machine - Principle and need of cardiopulmonary bypass - Operation and types of oxygenators (bubble, film and membrane type) - block diagram Describe the working of drug delivery systems, endoscopy systems and is an official topic in Module 3 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding machine Ventilators - terminologies - cycling mechanisms - modes of Operation - working of a typical ventilator (block diagram). Anaesthesia machine - principle - block diagram and working Heart lung machine - Principle and need of cardiopulmonary bypass - Operation and types of oxygenators (bubble, film and membrane type) - block diagram Describe the working of drug delivery systems, endoscopy systems and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding machine ventilators - terminologies - cycling mechanisms - modes of operation - working of a typical ventilator (block diagram). anaesthesia machine - principle - block diagram and working heart lung machine - principle and need of cardiopulmonary bypass - operation and types of oxygenators (bubble, film and membrane type) - block diagram describe the working of drug delivery systems, endoscopy systems and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding machine Ventilators - terminologies - cycling mechanisms - modes of Operation - working of a typical ventilator (block diagram). Anaesthesia machine - principle - block diagram and working Heart lung machine - Principle and need of cardiopulmonary bypass - Operation and types of oxygenators (bubble, film and membrane type) - block diagram Describe the working of drug delivery systems, endoscopy systems and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding machine Ventilators - terminologies - cycling mechanisms - modes of Operation - working of a typical ventilator (block diagram). Anaesthesia machine - principle - block diagram and working Heart lung machine - Principle and need of cardiopulmonary bypass - Operation and types of oxygenators (bubble, film and membrane type) - block diagram Describe the working of drug delivery systems, endoscopy systems and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

radiotherapy unit.

This module is presented from the official syllabus scope for Therapeutic Equipment. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: radiotherapy unit.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 5 Understanding infusion systems and their applications

5 Understanding infusion systems and their applications is an official topic in Module 4 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding infusion systems and their applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding infusion systems and their applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding infusion systems and their applications means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 5 Understanding infusion systems and their applications [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ State the principle of endoscopy.

State the principle of endoscopy. is an official topic in Module 4 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify State the principle of endoscopy. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, state the principle of endoscopy. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What State the principle of endoscopy. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ State the principle of endoscopy. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ Describe the parts of an endoscope 2 Understanding Explain the principle and working of Co-60

Describe the parts of an endoscope 2 Understanding Explain the principle and working of Co-60 is an official topic in Module 4 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the parts of an endoscope 2 Understanding Explain the principle and working of Co-60 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the parts of an endoscope 2 understanding explain the principle and working of co-60 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Describe the parts of an endoscope 2 Understanding Explain the principle and working of Co-60 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Describe the parts of an endoscope 2 Understanding Explain the principle and working of Co-60 □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 6 Understanding radiotherapy unit and LINACC system Principle of various types of drug infusion systems - Working and applications of syringe infusion pump - peristaltic infusion pump (block diagram) Endoscopy- principle and working with diagram- Parts of an endoscope with schematic- Types of endoscopy with applications. Radiotherapy - Isodose chart and biological effects of radiotherapy- Principle and operation of Co-60 radiotherapy unit, LINACC- principle and operation (with diagram), Applications of radiotherapy Text / Reference: T/R Book Title/Author R S Khandpur - Handbook of Biomedical instrumentation- third edition, Tata T1 McGraw Hill-India-2014 John G. Webster - Encyclopedia of Medical Devices and Instrumentation(6 R1 Volume set) , Wiley,7 April 2006 Joseph J Carr and John M Brown - Introduction to Biomedical Equipment R2 Technology- Fourth edition, Pearson Education India, 2008 Mandeep Singh -Introduction to Biomedical Instrumentation, Second Edition, R3 PHI Learning Pvt. Ltd., 01-Aug-2014 R S Khandpur- Handbook of Biomedical instrumentation- third edition, Tata R4 McGraw Hill-India-2014 Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=gk_Qf-JAL84 2 http://nptel.iitm.ac.in/ 3 http://www.nptel.ac.in/ 4 http://www.textbooksonline/

6 Understanding radiotherapy unit and LINACC system Principle of various types of drug infusion systems - Working and applications of syringe infusion pump - peristaltic infusion pump (block diagram) Endoscopy- principle and working with diagram- Parts of an endoscope with schematic- Types of endoscopy with applications. Radiotherapy - Isodose chart and biological effects of radiotherapy- Principle and operation of Co-60 radiotherapy unit, LINACC- principle and operation (with diagram), Applications of radiotherapy Text / Reference: T/R Book Title/Author R S Khandpur - Handbook of Biomedical instrumentation- third edition, Tata T1 McGraw Hill-India-2014 John G. Webster - Encyclopedia of Medical Devices and Instrumentation(6 R1 Volume set) , Wiley,7 April 2006 Joseph J Carr and John M Brown - Introduction to Biomedical Equipment R2 Technology- Fourth edition, Pearson Education India, 2008 Mandeep Singh -Introduction

to Biomedical Instrumentation, Second Edition, R3 PHI Learning Pvt. Ltd., 01-Aug-2014 R S Khandpur- Handbook of Biomedical instrumentation- third edition, Tata R4 McGraw Hill-India-2014 Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=gk_Qf-JAL84 2 http://nptel.iitm.ac.in/ 3 http://www.nptel.ac.in/ 4 http://www.textbooksonline/ is an official topic in Module 4 of Therapeutic Equipment. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding radiotherapy unit and LINACC system Principle of various types of drug infusion systems - Working and applications of syringe infusion pump - peristaltic infusion pump (block diagram) Endoscopy- principle and working with diagram- Parts of an endoscope with schematic- Types of endoscopy with applications. Radiotherapy - Isodose chart and biological effects of radiotherapy- Principle and operation of Co-60 radiotherapy unit, LINACC- principle and operation (with diagram), Applications of radiotherapy Text / Reference: T/R Book Title/Author R S Khandpur - Handbook of Biomedical instrumentation- third edition, Tata T1 McGraw Hill-India-2014 John G. Webster - Encyclopedia of Medical Devices and Instrumentation(6 R1 Volume set) , Wiley,7 April 2006 Joseph J Carr and John M Brown - Introduction to Biomedical Equipment R2 Technology- Fourth edition, Pearson Education India, 2008 Mandeep Singh - Introduction to Biomedical Instrumentation, Second Edition, R3 PHI Learning Pvt. Ltd., 01-Aug-2014 R S Khandpur- Handbook of Biomedical instrumentation- third edition, Tata R4 McGraw Hill-India-2014 Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=gk_Qf-JAL84 2 http://nptel.iitm.ac.in/ 3 http://www.nptel.ac.in/ 4 http://www.textbooksonline/ using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding radiotherapy unit and linacc system principle of various types of drug infusion systems - working and applications of syringe infusion pump - peristaltic infusion pump (block diagram) endoscopy- principle and working with diagram- parts of an endoscope with schematic- types of endoscopy with applications. radiotherapy - isodose chart and biological effects of radiotherapy- principle and operation of co-60 radiotherapy unit, linacc- principle and operation (with diagram), applications of radiotherapy text / reference: t/r book title/author r s khandpur - handbook of biomedical instrumentation- third edition, tata t1 mcgraw hill-india-2014 john g. webster - encyclopedia of medical devices and instrumentation(6 r1 volume set) , wiley,7 april 2006 joseph j carr and john m brown - introduction to biomedical equipment r2 technology- fourth edition, pearson education india, 2008 mandeep singh - introduction to biomedical instrumentation, second edition, r3 phi learning pvt. ltd., 01-aug-2014 r s khandpur- handbook of biomedical instrumentation- third edition, tata r4 mcgraw hill-india-2014 online resources: sl.no website link 1 https://www.youtube.com/watch?v=gk_qf-jal84 2 http://nptel.iitm.ac.in/ 3 http://www.nptel.ac.in/ 4 http://www.textbooksonline/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding radiotherapy unit and LINACC system Principle of various types of drug infusion systems - Working and applications of syringe infusion pump - peristaltic infusion pump (block diagram) Endoscopy- principle and working with diagram- Parts of an endoscope with schematic- Types of endoscopy with applications. Radiotherapy - Isodose chart and biological effects of radiotherapy- Principle and operation of Co-60 radiotherapy unit, LINACC- principle and operation (

Aspect	What to prepare
	with diagram), Applications of radiotherapy Text / Reference: T/R Book Title/Author R S Khandpur - Handbook of Biomedical instrumentation- third edition, Tata T1 McGraw Hill-India-2014 John G. Webster - Encyclopedia of Medical Devices and Instrumentation(6 R1 Volume set) , Wiley,7 April 2006 Joseph J Carr and John M Brown - Introduction to Biomedical Equipment R2 Technology- Fourth edition, Pearson Education India, 2008 Mandeep Singh -Introduction to Biomedical Instrumentation, Second Edition, R3 PHI Learning Pvt. Ltd., 01-Aug-2014 R S Khandpur- Handbook of Biomedical instrumentation- third edition, Tata R4 McGraw Hill-India-2014 Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=gk_Qf-JAL84 2 http://nptel.iitm.ac.in/ 3 http://www.nptel.ac.in/ 4 http://www.textbooksonline/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 6 Understanding radiotherapy unit and LINACC system Principle of various types of drug infusion systems - Working and applications of syringe infusion pump - peristaltic infusion pump (block diagram) Endoscopy- principle and working with diagram- Parts of an endoscope with schematic- Types of endoscopy with applications. Radiotherapy - Isodose chart and biological effects of radiotherapy- Principle and operation of Co-60 radiotherapy unit, LINACC- principle and operation (with diagram), Applications of radiotherapy Text / Reference: T/R Book Title/Author R S Khandpur - Handbook of Biomedical instrumentation- third edition, Tata T1 McGraw Hill-India-2014 John G. Webster - Encyclopedia of

Medical Devices and Instrumentation(6 R1 Volume set) , Wiley,7 April 2006 Joseph J Carr and John M Brown - Introduction to Biomedical Equipment R2 Technology- Fourth edition, Pearson Education India, 2008 Mandeep Singh - Introduction to Biomedical Instrumentation, Second Edition, R3 PHI Learning Pvt. Ltd., 01-Aug-2014 R S Khandpur- Handbook of Biomedical instrumentation- third edition, Tata R4 McGraw Hill-India-2014 Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=gk_Qf-JAL84 2 http://nptel.iitm.ac.in/ 3 http://www.nptel.ac.in/ 4 http://www.textbooksonline/ □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 6 Understanding principle and working Identify the types of defibrillators, their. (3 marks)**

▶ **Define or explain 6 Understanding principle and working. Explain pain relief through electrical stimulation. (3 marks)**

▶ **Define or explain 3 Understanding and types of electrical stimulators. Cardiac pacemakers - design requirements - Power sources - Pacing modes - Working of ventricular synchronous demand pacemaker - Programmable pacemaker (block diagram and description). Defibrillators - Principle - working and output waveforms of DC defibrillators - Defibrillator with synchronizer - cardioverter -Automatic Implantable Defibrillators (block diagram and working) Electrical stimulators - Principle - different types of electrical stimulators - TENS- Spinal Cord Stimulator - Magnetic stimulation Explain the principle and working of instruments for surgery, Dialysis and. (3 marks)**

Module 2

▶ **Define or explain 4 Understanding diathermy unit. Classify Lithotripters based on their principle. (3 marks)**

▶ **Define or explain and working. Describe dialysis and the working of. (3 marks)**

▶ **Define or explain 4 Understanding heamodialyzers Surgical diathermy -Working principle - modes of operation(schematic diagram and waveforms) - block diagram and working of ESU Lithotripsy - Principle - operation of extra corporeal shock wave lithotripter and ultrasonic lithotripter with schematic Dialysis - Principle - types of dialyzers (parallel flow, hollow fibre and coil type) - heamodialysis machine principle and working with block diagram Summarize the principle and operation of the ICU and operation theatre. (3 marks)**

Module 3

▶ **Define or explain 6 Understanding Respiration and the types of ventilators. Explain the principle and working of. (3 marks)**

▶ **Define or explain anesthesia machine. (3 marks)**

▶ **Define or explain Identify the need for cardiopulmonary bypass. 2 Understanding Describe the principle working of Heart lung. (3 marks)**

▶ **Define or explain 3 Understanding machine Ventilators - terminologies - cycling mechanisms - modes of Operation - working of a typical ventilator (block diagram). Anaesthesia machine - principle - block diagram and working Heart lung machine - Principle and need of cardiopulmonary bypass - Operation and types of oxygenators (bubble, film and membrane type) - block diagram Describe the working of drug delivery systems, endoscopy systems and. (3 marks)**

Module 4

▶ **Define or explain 5 Understanding infusion systems and their applications. (3 marks)**

▶ **Define or explain State the principle of endoscopy.. (3 marks)**

▶ **Define or explain Describe the parts of an endoscope 2 Understanding Explain the principle and working of Co-60. (3 marks)**

▶ **Define or explain 6 Understanding radiotherapy unit and LINACC system Principle of various types of drug infusion systems - Working and applications of syringe infusion pump - peristaltic infusion pump (block diagram) Endoscopy- principle and working with diagram- Parts of an endoscope with schematic- Types of endoscopy with applications. Radiotherapy - Isodose chart and biological effects of radiotherapy- Principle and operation of Co-60 radiotherapy unit, LINACC- principle and operation (with diagram), Applications of radiotherapy Text / Reference: T/R Book Title/Author R S Khandpur - Handbook of Biomedical instrumentation- third edition, Tata T1 McGraw Hill-India-2014 John G. Webster - Encyclopedia of Medical Devices and Instrumentation(6 R1 Volume set) , Wiley,7 April 2006 Joseph J Carr and John M Brown - Introduction to Biomedical Equipment R2 Technology-Fourth edition, Pearson Education India, 2008 Mandeep Singh - Introduction to Biomedical Instrumentation, Second Edition, R3 PHI Learning Pvt. Ltd., 01-Aug-2014 R S Khandpur- Handbook of Biomedical instrumentation- third edition, Tata R4 McGraw Hill-India-2014 Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=gk_Qf-JAL84 2 http://nptel.iitm.ac.in/ 3 http://www.nptel.ac.in/ 4 http://www.textbooksonline/. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **6 Understanding principle and working Identify the types of defibrillators, their**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding diathermy unit. Classify Lithotripters based on their principle**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **6 Understanding Respiration and the types of ventilators. Explain the principle and working of**.

Module 4

1-mark definition: State the meaning of **5 Understanding infusion systems and their applications**.

3-mark explanation: Explain one principle, process or comparison from this module.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://www.youtube.com/watch?v=gk_Qf-JAL84 http://nptel.iitm.ac.in/ http://www.nptel.ac.in/ http://www.textbooksonline/

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 4241. Verify institutional updates against the current official curriculum.